Mixed Traffic Test (Serial)

2026-03-11-19-18-58

Test Setup Information

Overall Setup Info For all Tests	DUT Model	Test_DUT
	DUT Firmware	NA
	SSID	NETGEAR_2G_wpa2
	Security	wpa2
	No of Devices	4 (Virtual Clients: 4, Windows: 0, Linux: 0, Mac: 0, Android: 0 ,iOS: 0)
	Test Duration (HH:MM:SS)	00:13:00

Objective

The Candela mixed traffic test is designed to measure the access point performance andstability by running multiple traffic on real clients like Android, Linux, Windows, and IOSconnected to the access point. This test allows the user to choose multiple types of traffic likeclient capacity test, web browser test, video streaming test ping test. Along with theperformance measurements are client connection times, Station 4-Way Handshake time, DHCPtimes, and more. The expected behavior is for the AP to be able to handle all types of traffic onthe several stations (within the limitations of the AP specs) and Make sure all clients can run alltypes of traffic.

Traffic Details

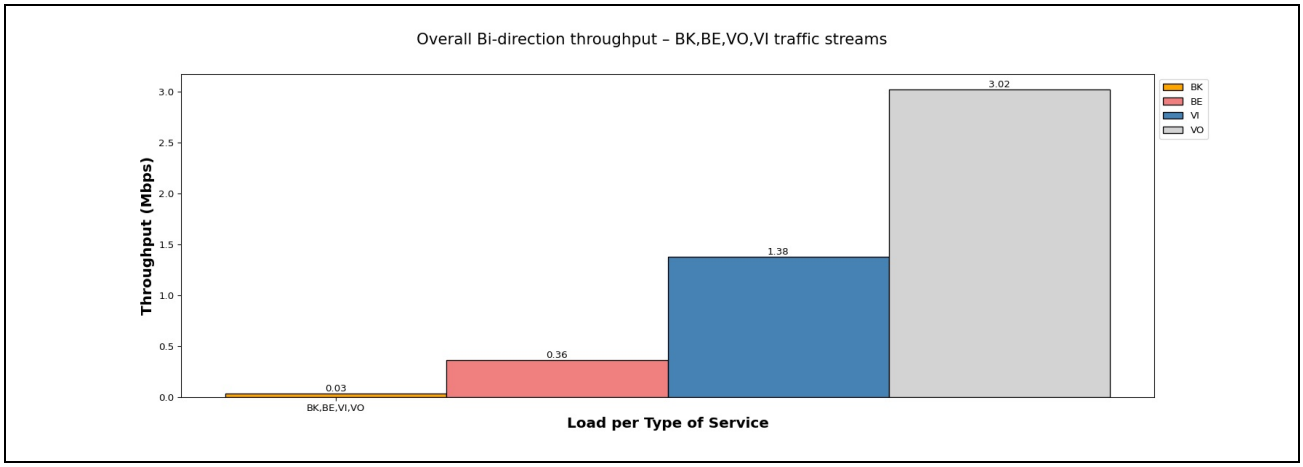
Sno	Test Cases	Test Duration	Test Status
1	Ping Test		
2	Quality Of Service(QOS) Test ['VI', 'VO', 'BK', 'BE']	1.0 minute	Executed
3	FTP Test		
4	HTTP Test		
5	Multicast Test		

2. Quality Of Service(QOS) Test

Test Configuration

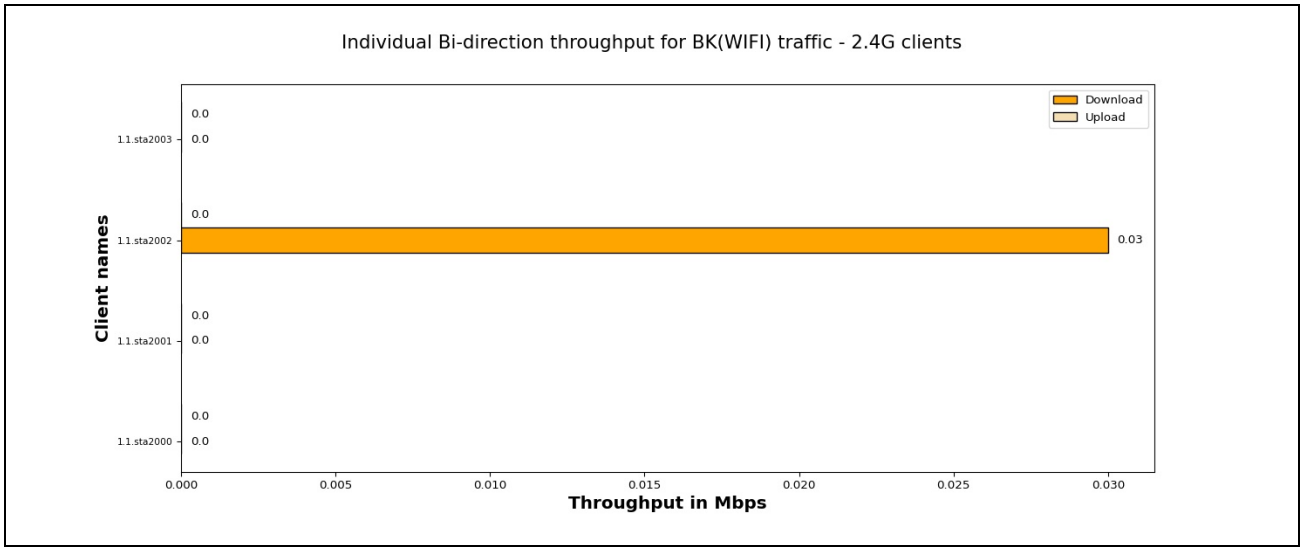
Test Setup Information	Protocol	TCP
	Traffic Direction	Bi-direction
	Security	wpa2
	TOS	['VI', 'VO', 'BK', 'BE']
	Per TOS Load	Upload:100.0 Mbps,Download:100.0 Mbps

Overall Bi-direction Throughput for all TOS i.e BK | BE | Video (VI) | Voice (VO)



Individual Bi-direction throughput with intended load Upload:100.0 Mbps,Download:100.0 Mbps/station for traffic BK(WiFi).

The below graph represents individual throughput for 4 clients running BK (WiFi) traffic. Y- axis shows "Client names" and X-axis shows "Throughput in Mbps".

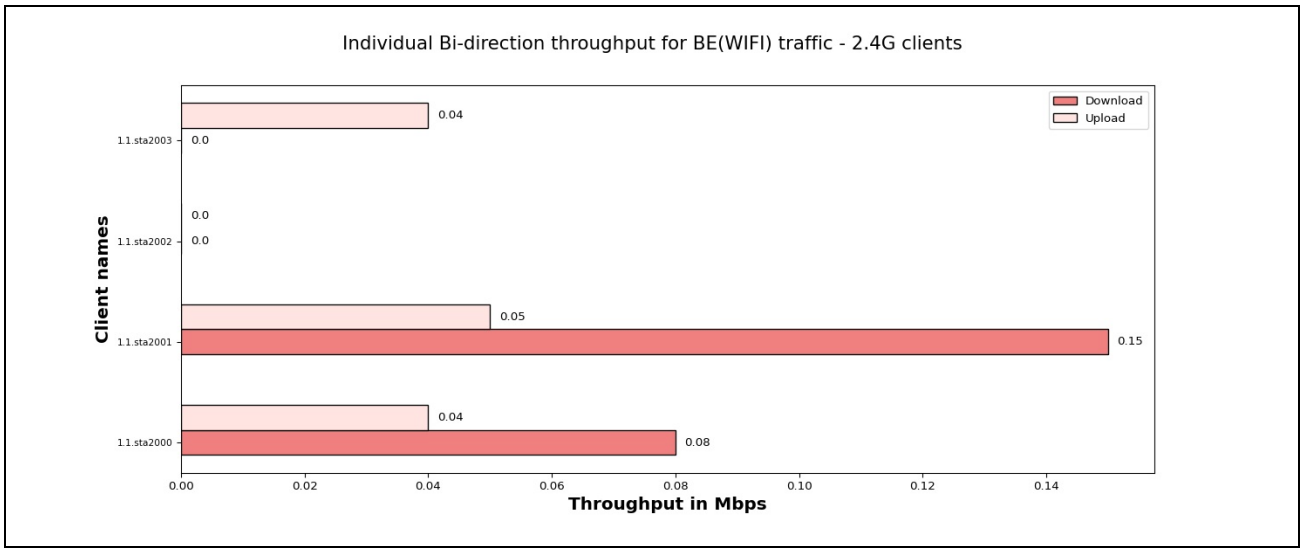


TOS : Background

Client Name	Mac	Channel	SSID	Type of traffic	Traffic Direction	Traffic Protocol	Offered upload rate(Mbps)	Offered download rate(Mbps)	Observed upload rate(Mbps)	Observed download rate(Mbps)	Observed Upload Drop (%)	Observed Download Drop (%)
1.1.sta2000	04:f0:21:94:eb:46	11	NETGEAR_2G_wpa2	Background	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.0	0.00	100.0	100.000000
1.1.sta2001	04:f0:21:94:90:46	11	NETGEAR_2G_wpa2	Background	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.0	0.00	100.0	100.000000
1.1.sta2002	04:f0:21:94:1f:46	11	NETGEAR_2G_wpa2	Background	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.0	0.03	100.0	66.666667
1.1.sta2003	04:f0:21:94:11:46	11	NETGEAR_2G_wpa2	Background	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.0	0.00	100.0	100.000000

Individual Bi-direction throughput with intended load Upload:100.0 Mbps,Download:100.0 Mbps/station for traffic BE(WiFi).

The below graph represents individual throughput for 4 clients running BE (WiFi) traffic. Y- axis shows "Client names" and X-axis shows "Throughput in Mbps".

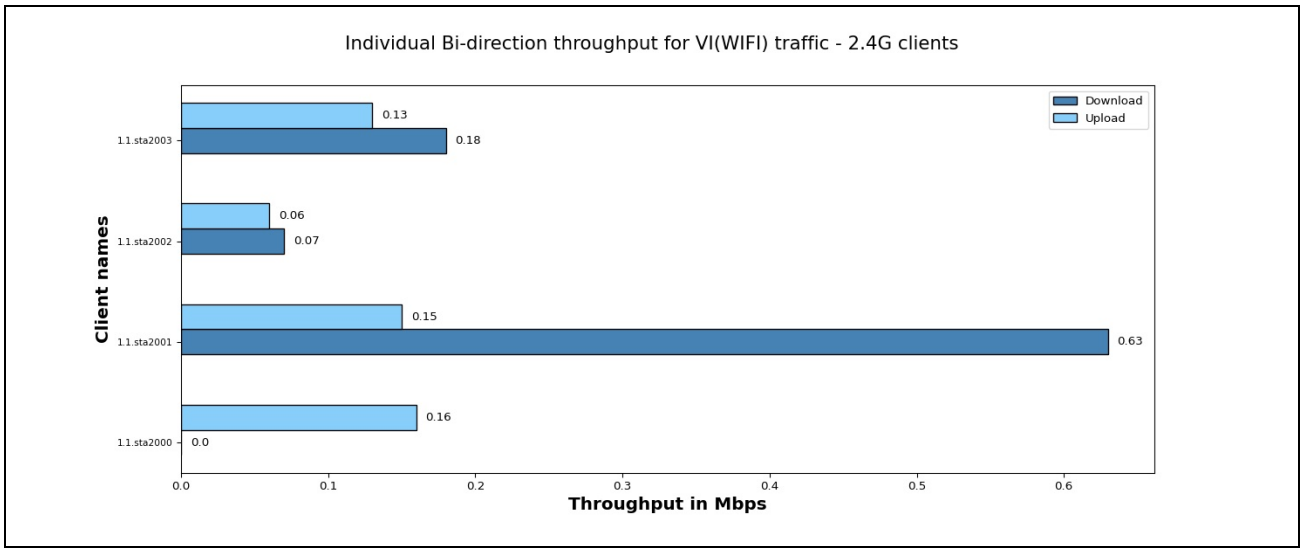


TOS : Besteffort

Client Name	Mac	Channel	SSID	Type of traffic	Traffic Direction	Traffic Protocol	Offered upload rate(Mbps)	Offered download rate(Mbps)	Observed upload rate(Mbps)	Observed download rate(Mbps)	Observed Upload Drop (%)	Observed Download Drop (%)
1.1.sta2000	04:f0:21:94:eb:46	11	NETGEAR_2G_wpa2	Besteffort	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.04	0.08	16.666667	50.000000
1.1.sta2001	04:f0:21:94:90:46	11	NETGEAR_2G_wpa2	Besteffort	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.05	0.15	100.000000	33.333333
1.1.sta2002	04:f0:21:94:1f:46	11	NETGEAR_2G_wpa2	Besteffort	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.00	0.00	100.000000	66.666667
1.1.sta2003	04:f0:21:94:11:46	11	NETGEAR_2G_wpa2	Besteffort	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.04	0.00	100.000000	66.666667

Individual Bi-direction throughput with intended load Upload:100.0 Mbps,Download:100.0 Mbps/station for traffic VI(WiFi).

The below graph represents individual throughput for 4 clients running VI (WiFi) traffic. Y- axis shows "Client names" and X-axis shows "Throughput in Mbps".

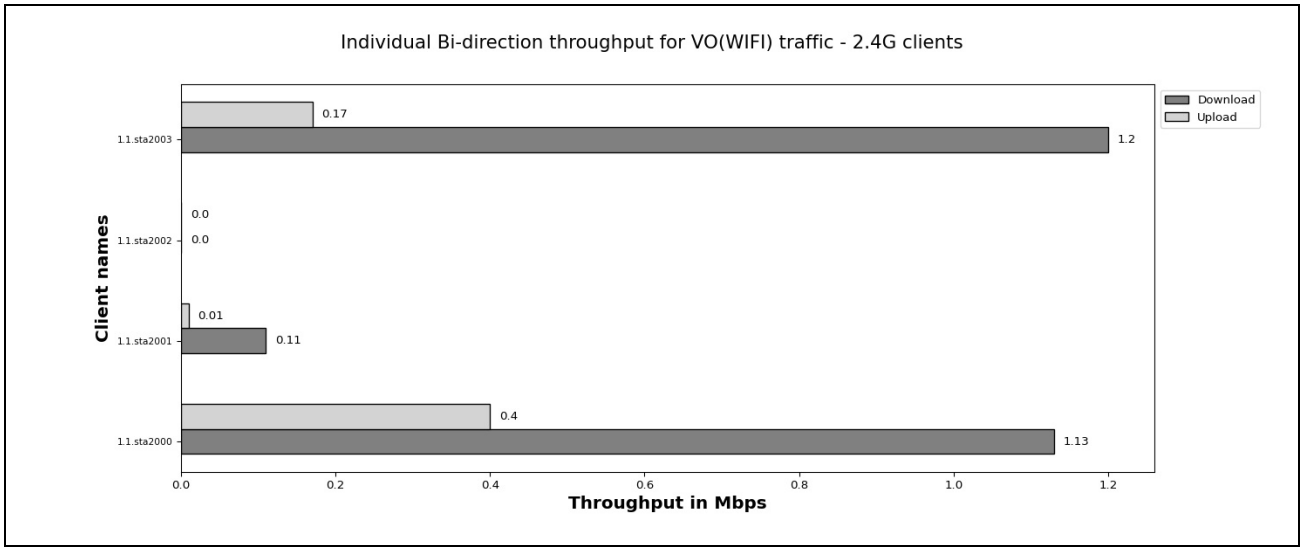


TOS : Video

Client Name	Mac	Channel	SSID	Type of traffic	Traffic Direction	Traffic Protocol	Offered upload rate(Mbps)	Offered download rate(Mbps)	Observed upload rate(Mbps)	Observed download rate(Mbps)	Observed Upload Drop (%)	Observed Download Drop (%)
1.1.sta2000	04:f0:21:94:eb:46	11	NETGEAR_2G_wpa2	Video	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.16	0.00	30.769231	30.769231
1.1.sta2001	04:f0:21:94:90:46	11	NETGEAR_2G_wpa2	Video	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.15	0.63	18.181818	20.000000
1.1.sta2002	04:f0:21:94:1f:46	11	NETGEAR_2G_wpa2	Video	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.06	0.07	23.076923	44.776119
1.1.sta2003	04:f0:21:94:11:46	11	NETGEAR_2G_wpa2	Video	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.13	0.18	16.666667	20.454545

Individual Bi-direction throughput with intended load Upload:100.0 Mbps,Download:100.0 Mbps/station for traffic VO(WiFi).

The below graph represents individual throughput for 4 clients running VO (WiFi) traffic. Y- axis shows "Client names" and X-axis shows "Throughput in Mbps".



TOS : Voice

Client Name	Mac	Channel	SSID	Type of traffic	Traffic Direction	Traffic Protocol	Offered upload rate(Mbps)	Offered download rate(Mbps)	Observed upload rate(Mbps)	Observed download rate(Mbps)	Observed Upload Drop (%)	Observed Download Drop (%)
1.1.sta2000	04:f0:21:94:eb:46	11	NETGEAR_2G_wpa2	Voice	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.40	1.13	13.461538	2.185792
1.1.sta2001	04:f0:21:94:90:46	11	NETGEAR_2G_wpa2	Voice	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.01	0.11	100.000000	32.786885
1.1.sta2002	04:f0:21:94:1f:46	11	NETGEAR_2G_wpa2	Voice	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.00	0.00	100.000000	29.729730

1.1.sta2003	04:f0:21:94:11:46	11	NETGEAR_2G_wpa2	Voice	Bi-direction	TCP	100.0 Mbps	100.0 Mbps	0.17	1.20	21.428571	8.284024

Overall Info	contact	support@candelatech.com